

# Task 1

K9

2025-12-09

Loading librarys:

```
#install.packages("tidyverse")  
#install.packages("ggplot2")
```

```
library(tidyverse)
```

```
## -- Attaching core tidyverse packages ----- tidyverse 2.0.0 --  
## v dplyr      1.1.4      v readr      2.1.5  
## v forcats    1.0.1      v stringr   1.5.2  
## v ggplot2     4.0.0      v tibble    3.3.0  
## v lubridate  1.9.4      v tidyr     1.3.1  
## v purrr      1.1.0  
## -- Conflicts ----- tidyverse_conflicts() --  
## x dplyr::filter() masks stats::filter()  
## x dplyr::lag()     masks stats::lag()  
## i Use the conflicted package (<http://conflicted.r-lib.org/>) to force all conflicts to become errors
```

```
library(ggplot2)
```

Import data

```
gdp_per_capita_df <- read.csv("Exported_data/gdp_per_capita_cleaned.csv")  
gdp_per_capita_df <- gdp_per_capita_df %>%  
  filter(Year >= 2015)
```

Calculate the total GDP of each continent

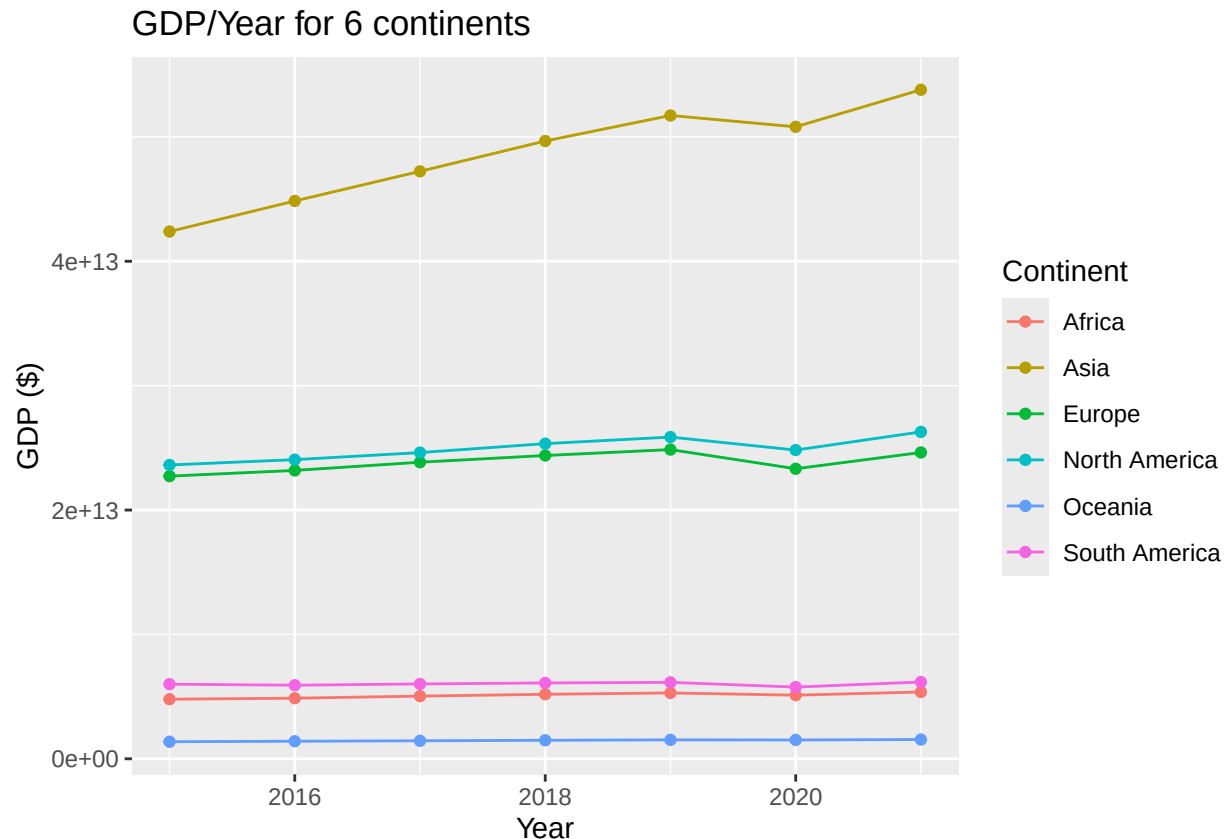
```
continent_gdp_totals <- gdp_per_capita_df %>%  
  group_by(Continent, Year) %>%  
  summarise(  
    Total_GDP = sum(Gdp, na.rm = TRUE)  
  ) %>%  
  ungroup()
```

```
## 'summarise()' has grouped output by 'Continent'. You can override using the  
## '.groups' argument.
```

## Target 1.1 - Sustain per capita economic growth in accordance with national circumstances

Plot a graph of GDP against time for each continent

```
continent_gdp_totals %>% ggplot(aes(x = Year, y = Total_GDP, colour = Continent)) +  
  geom_point() +  
  geom_line() +  
  labs(title = "GDP/Year for 6 continents", y = "GDP ($)")
```



I decided this graph could show more if they all started at an index of 100

```
continent_gdp_totals <- continent_gdp_totals %>%  
  group_by(Continent) %>%  
  mutate(  
    GDP_Index = Total_GDP / first(Total_GDP) * 100  
  )
```

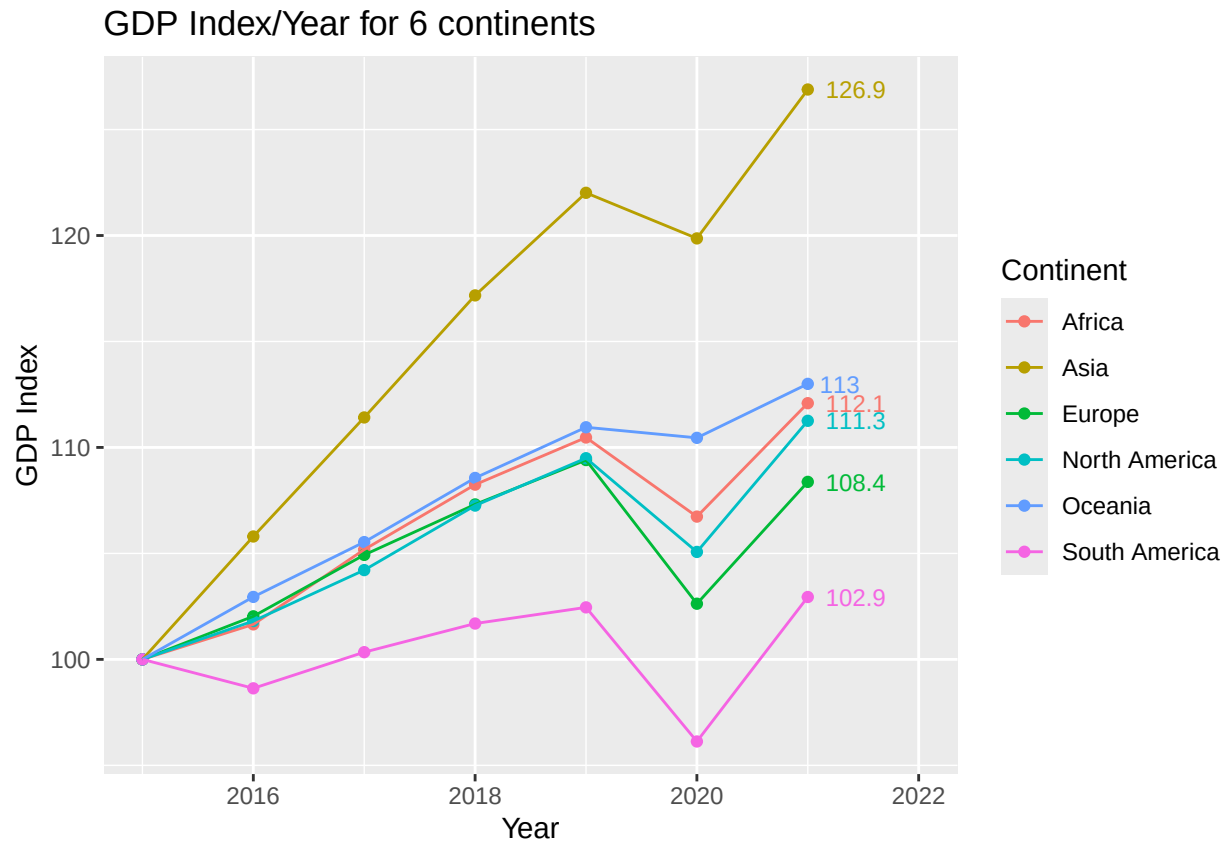
We now replot the graph, using indexed values (starting value index 100)

```
continent_gdp_totals %>% ggplot(aes(x = Year, y = GDP_Index, colour = Continent)) +  
  geom_point() +  
  geom_line() +  
  geom_text(  
    data = continent_gdp_totals %>% filter(Year == 2021),
```

```

aes(label = round(GDP_Index, 1)),
hjust = -0.3,
size = 3,
show.legend = FALSE) +
expand_limits(x = 2022) +
labs(title = "GDP Index/Year for 6 continents", y = "GDP Index")

```



Calculate the weighted and mean growth of each continent

```

continents_growth_df <- gdp_per_capita_df %>%
  group_by(Continent, Year) %>%
  summarise(
    Weighted_Growth = sum(Growth * Gdp, na.rm = TRUE) / sum(Gdp, na.rm = TRUE),
    Avg_Growth = mean(Growth, na.rm = TRUE)
  ) %>%
  ungroup()

```

## 'summarise()' has grouped output by 'Continent'. You can override using the  
## '.groups' argument.

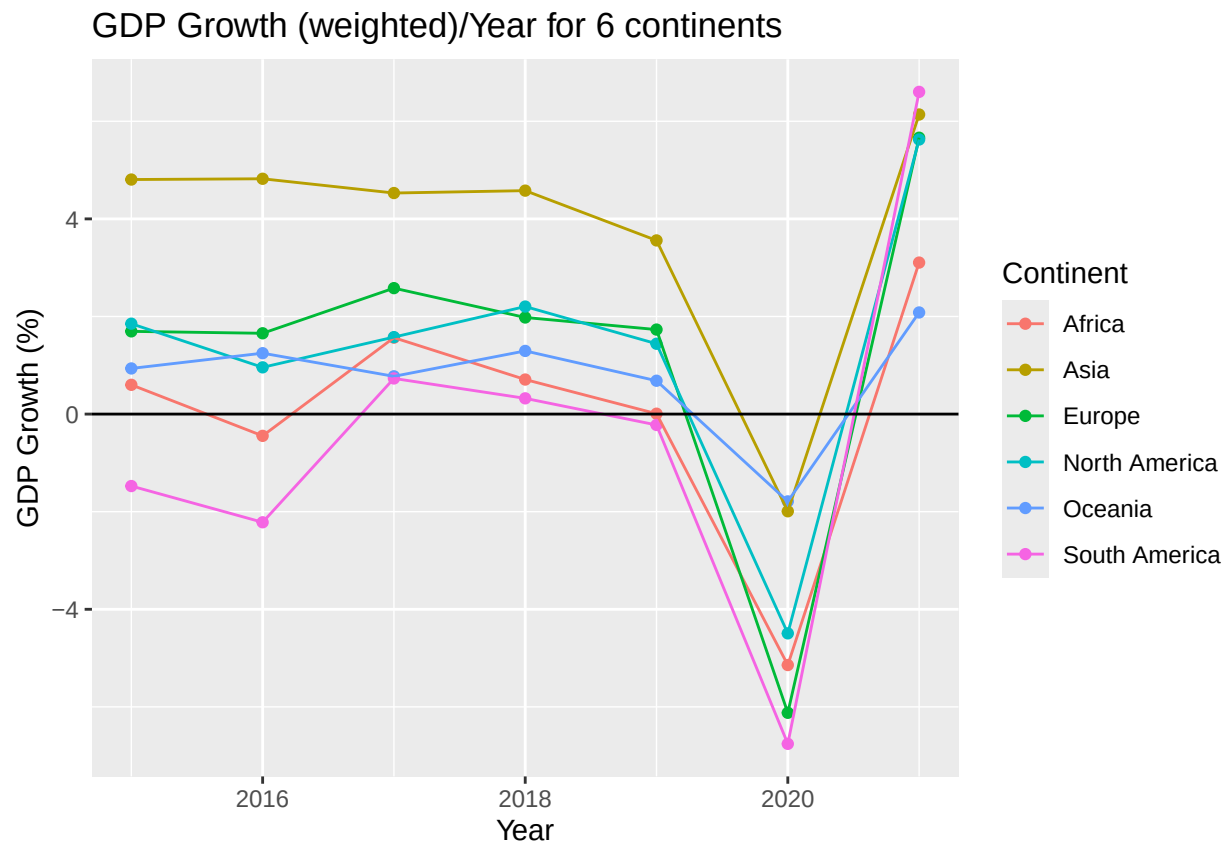
Plot the growth graphs of each continent

```

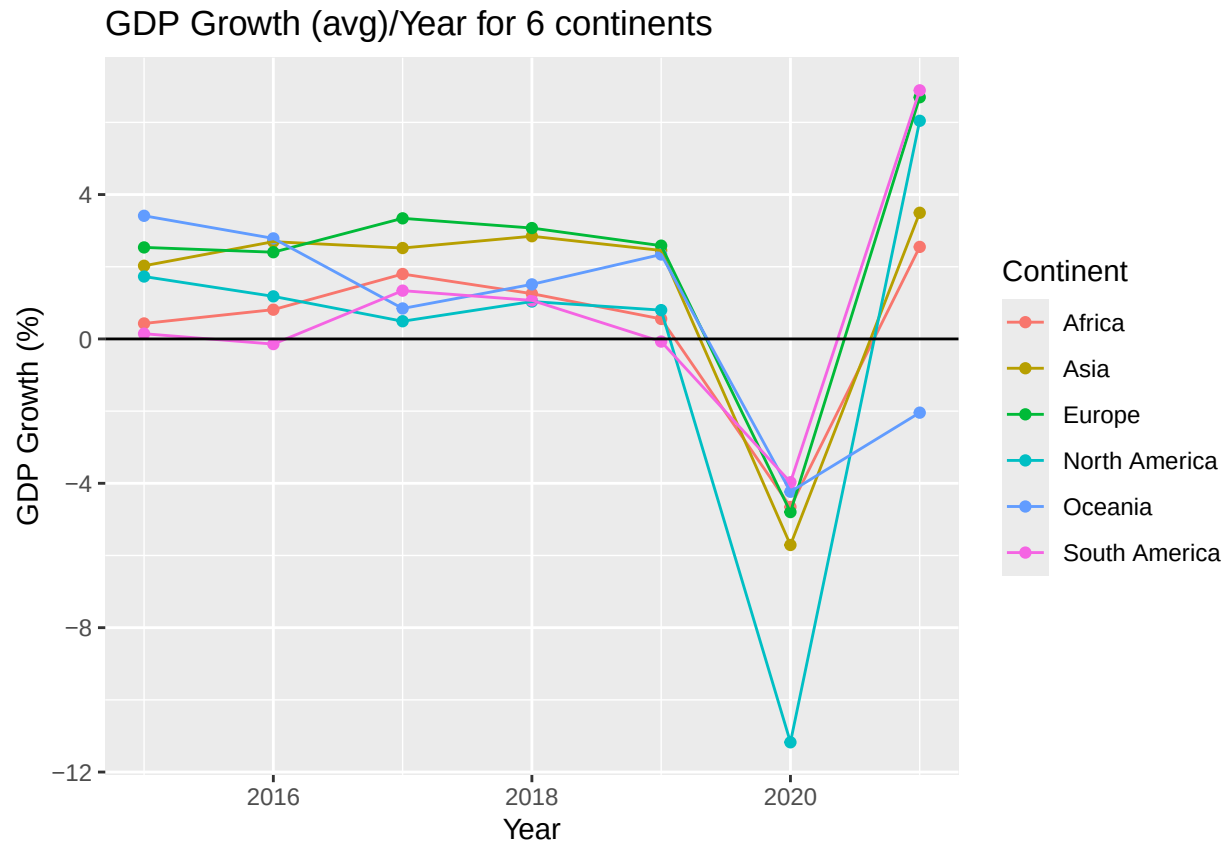
continents_growth_df %>% ggplot(aes(x = Year, y = Weighted_Growth, colour = Continent)) +
  geom_point() +

```

```
geom_line() +
geom_hline(yintercept = 0) +
labs(title = "GDP Growth (weighted)/Year for 6 continents", y = "GDP Growth (%)")
```



```
continents_growth_df %>% ggplot(aes(x = Year, y = Avg_Growth, colour = Continent)) +
  geom_point() +
  geom_line() +
  geom_hline(yintercept = 0) +
  labs(title = "GDP Growth (avg)/Year for 6 continents", y = "GDP Growth (%)")
```

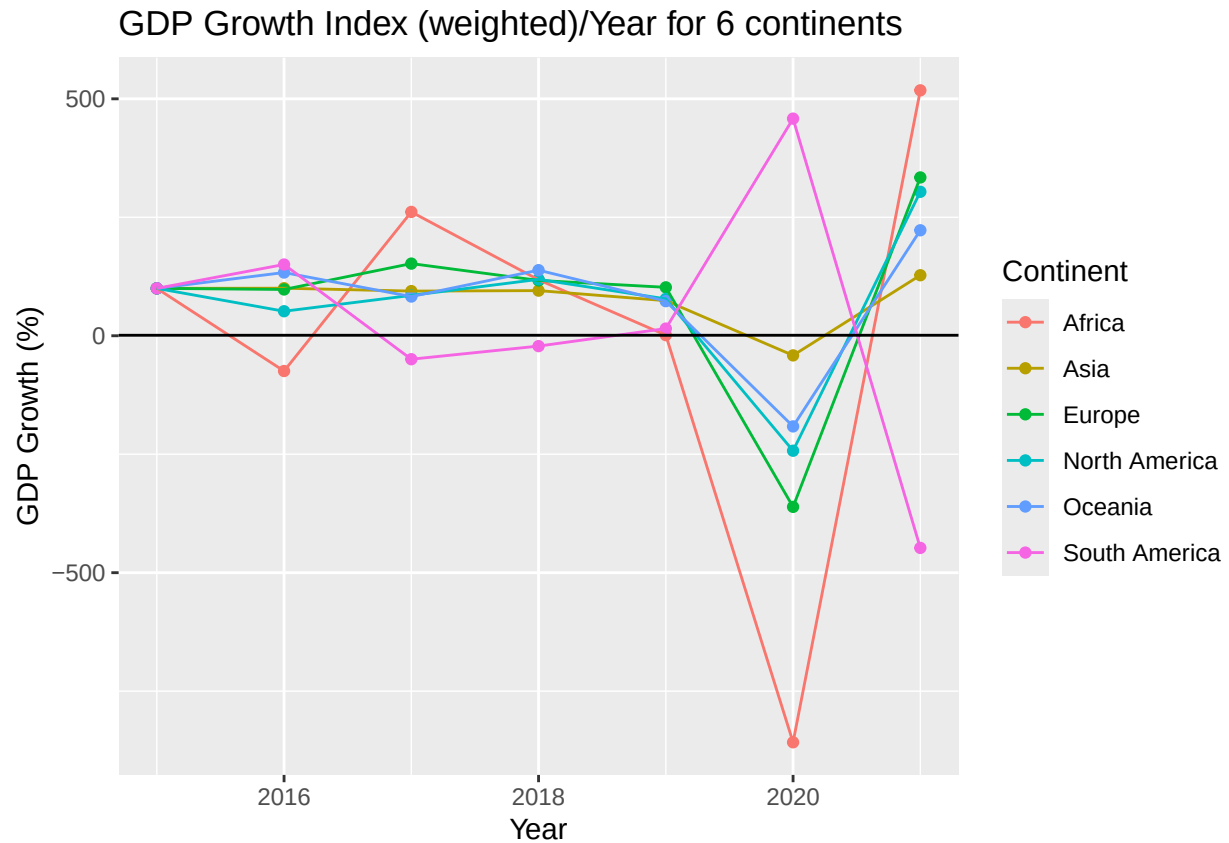


I decided to do a similar thing with growth about index (realised is redundant, don't use in analysis)

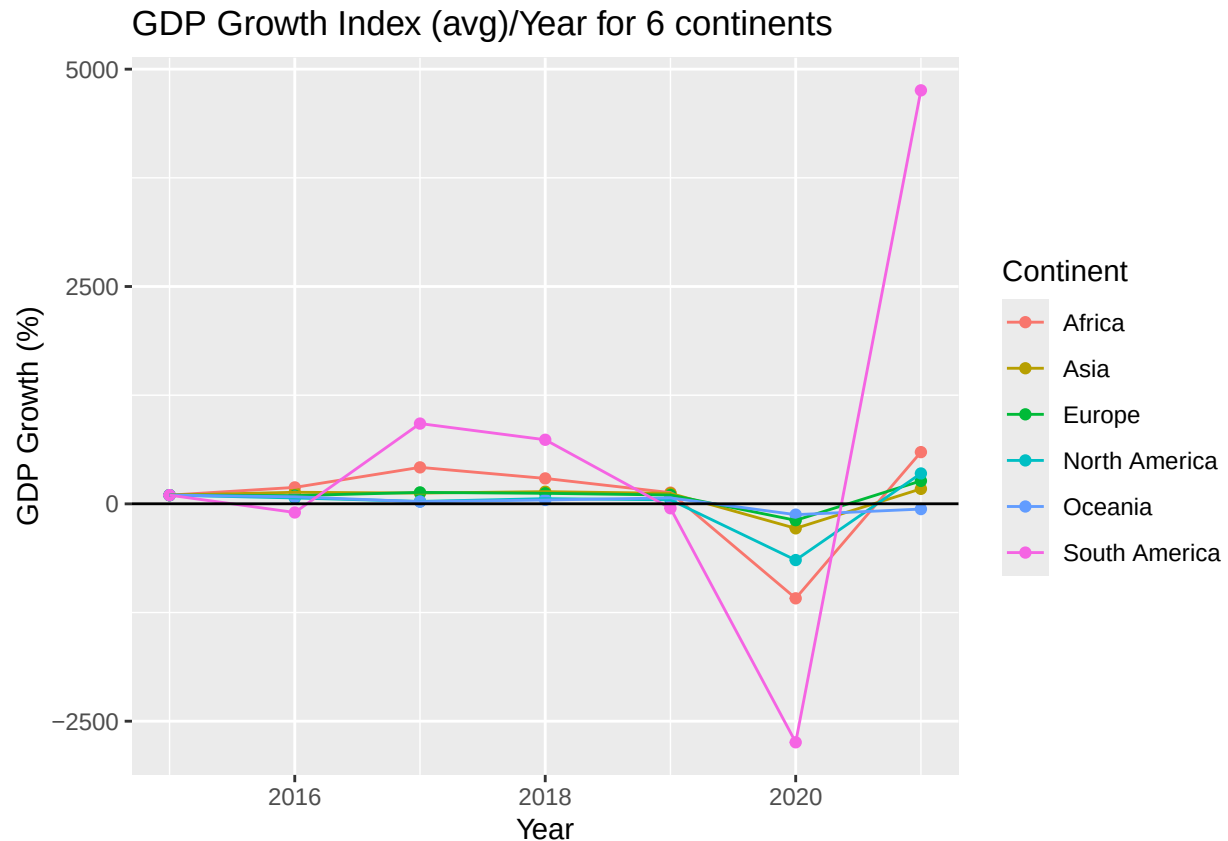
```
continents_growth_df <- continents_growth_df %>%
  group_by(Continent) %>%
  mutate(
    Avg_Growth_Index = Avg_Growth / first(Avg_Growth) * 100,
    Weighted_Growth_Index = Weighted_Growth / first(Weighted_Growth) * 100
  )
```

Plotting the graphs

```
continents_growth_df %>% ggplot(aes(x = Year, y = Weighted_Growth_Index, colour = Continent)) +
  geom_point() +
  geom_line() +
  geom_hline(yintercept = 1) +
  labs(title = "GDP Growth Index (weighted)/Year for 6 continents", y = "GDP Growth (%)")
```



```
continents_growth_df %>% ggplot(aes(x = Year, y = Avg_Growth_Index, colour = Continent)) +
  geom_point() +
  geom_line() +
  geom_hline(yintercept = 1) +
  labs(title = "GDP Growth Index (avg)/Year for 6 continents", y = "GDP Growth (%)")
```



Now we'll plot a graph to see if top 2 GDP countries are the drivers between continental. Making a new data frame with the top 2 countries per continent and their gdp's

```
top_2_per_continent <- gdp_per_capita_df %>%
  group_by(Continent, Entity) %>%
  summarise(Total_GDP = sum(Gdp, na.rm = TRUE)) %>%
  arrange(Continent, desc(Total_GDP)) %>%
  group_by(Continent) %>%
  slice(1:2) %>%
  ungroup()
```

## 'summarise()' has grouped output by 'Continent'. You can override using the  
## '.groups' argument.

Function which returns a dataset containing GDP growth for the top 2 countries in a continent and the continent's average growth

```
get_plot_data <- function(x) {

  top_2_entities <- top_2_per_continent %>%
    filter(Continent == x) %>%
    pull(Entity)

  country_data <- gdp_per_capita_df %>%
    filter(Continent == x, Entity %in% top_2_entities)
```

```

continent_avg <- continents_growth_df %>%
  filter(Continent == x) %>%
  rename(Growth = Avg_Growth) %>%
  mutate(Entity = "Continent Average")

plot_data <- bind_rows(country_data, continent_avg)

return(plot_data)
}

```

Function to plot the graphs

```

plot_drivers <- function(continent) {
  data <- get_plot_data(continent)

  ggplot(data, aes(x = Year, y = Growth, colour = Entity)) +
    geom_line() +
    geom_point() +
    labs(
      title = paste("Top 2 GDP Growth vs Avg -", continent),
      y = "Growth (%)"
    )
}

```

Plotting the graphs

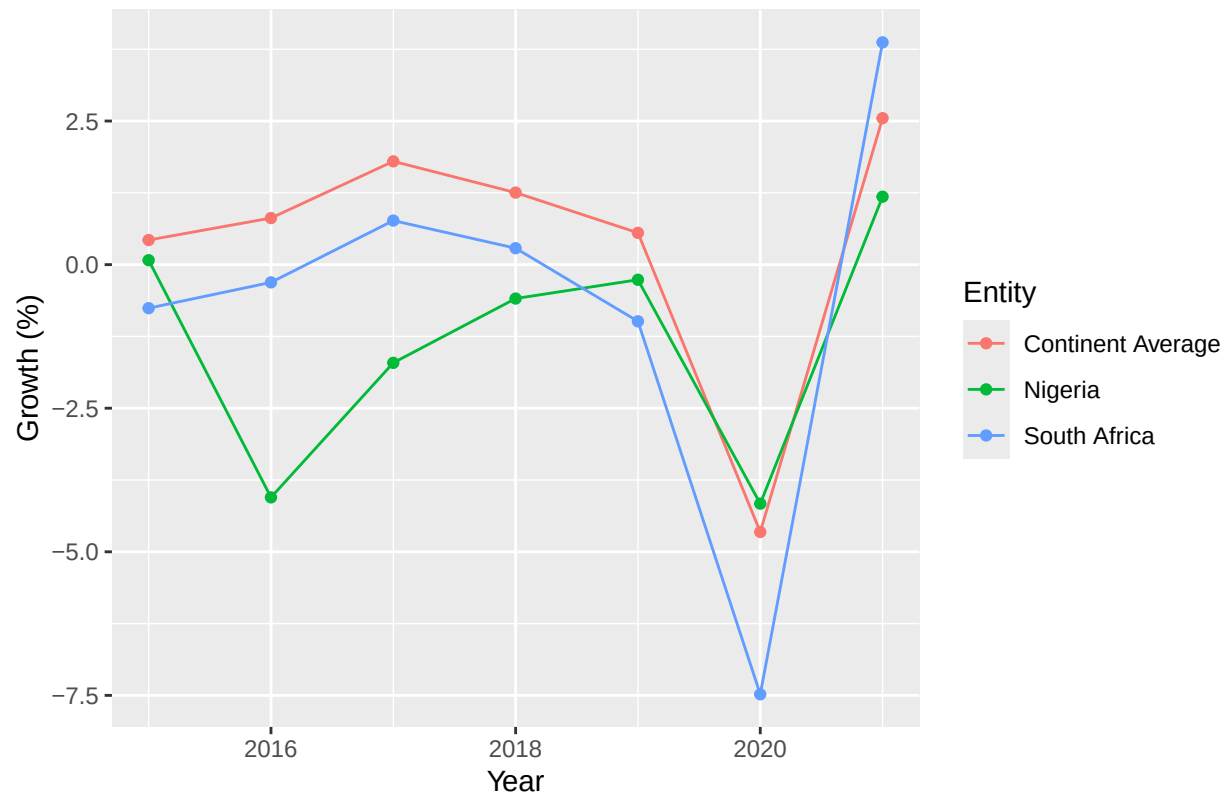
```

plot_drivers("Africa")

```

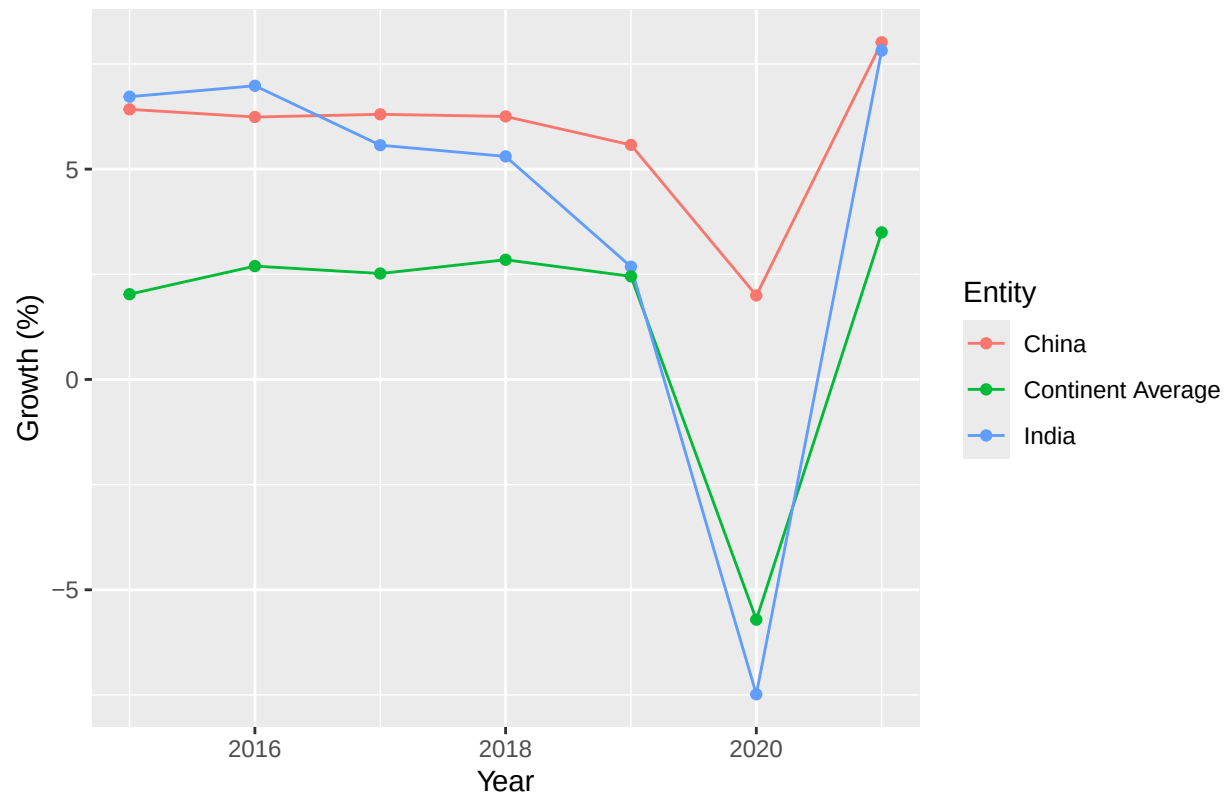


Top 2 GDP Growth vs Avg – Africa



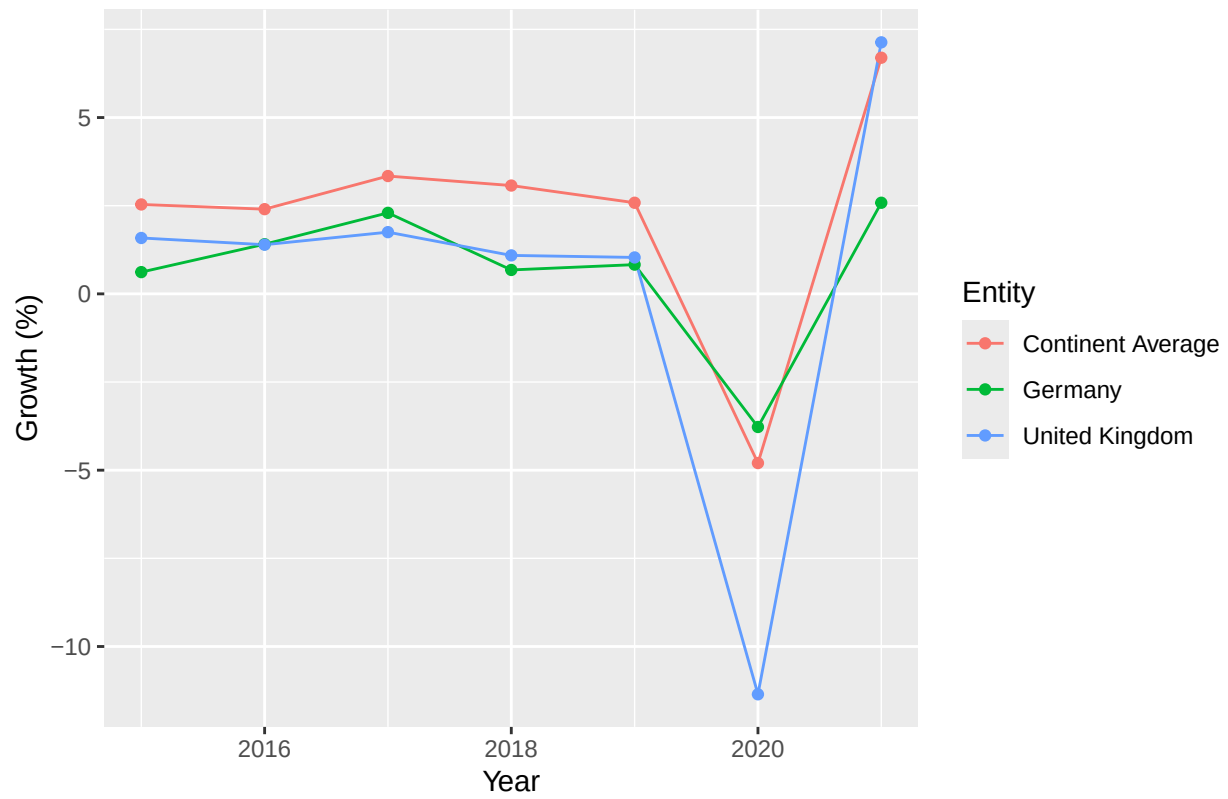
```
plot_drivers("Asia")
```

Top 2 GDP Growth vs Avg – Asia



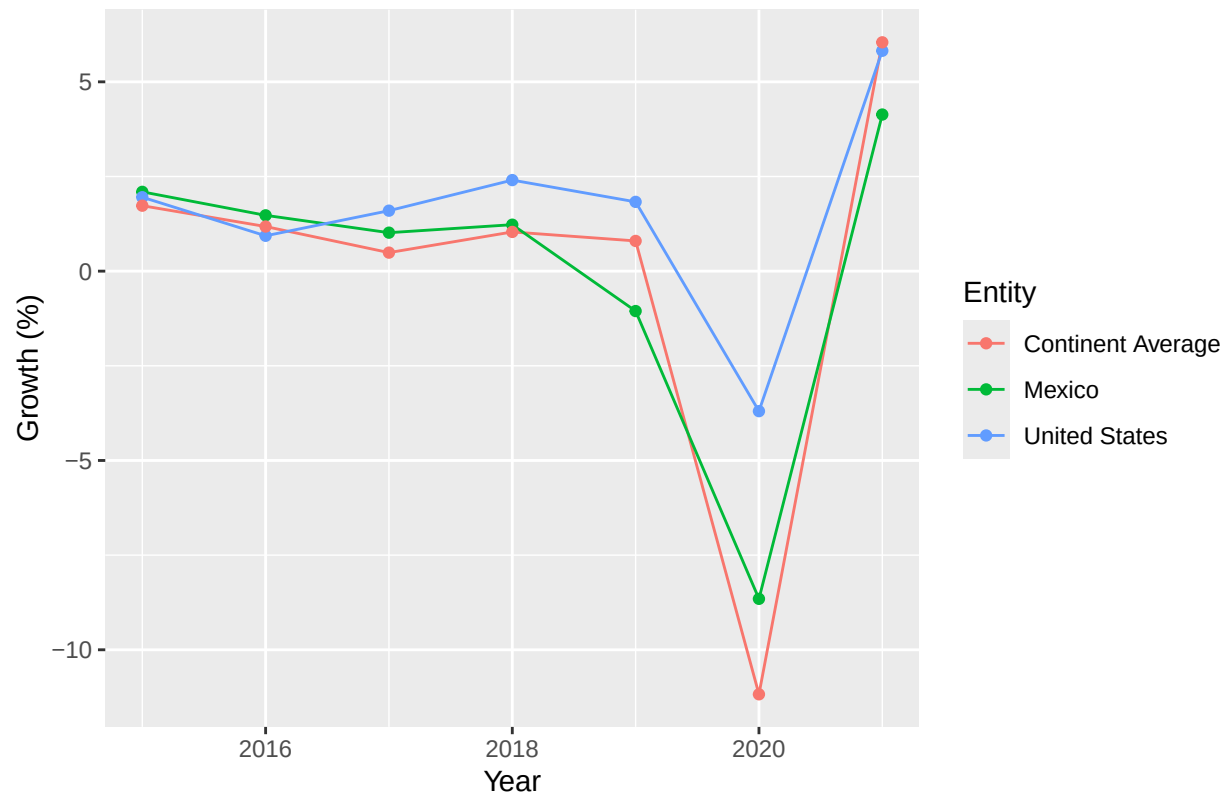
```
plot_drivers("Europe")
```

Top 2 GDP Growth vs Avg – Europe

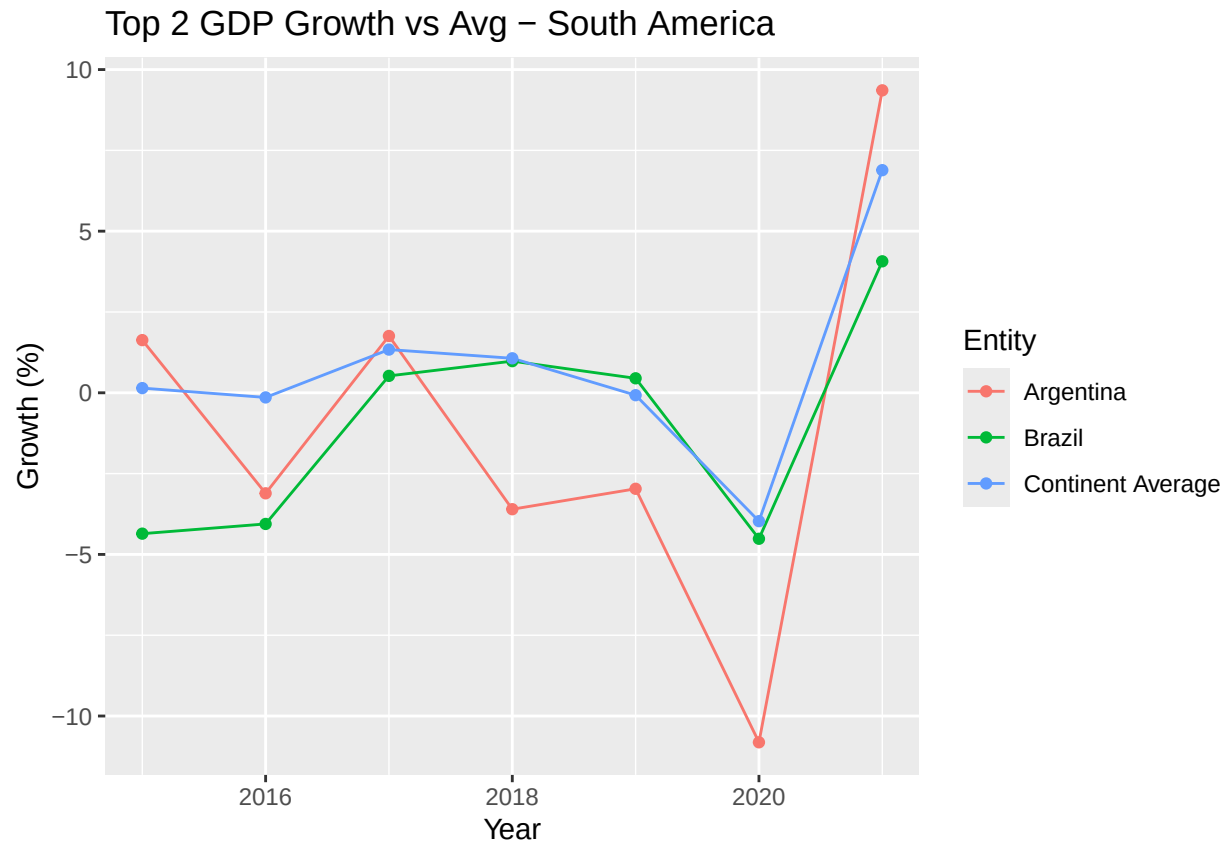


```
plot_drivers("North America")
```

Top 2 GDP Growth vs Avg – North America

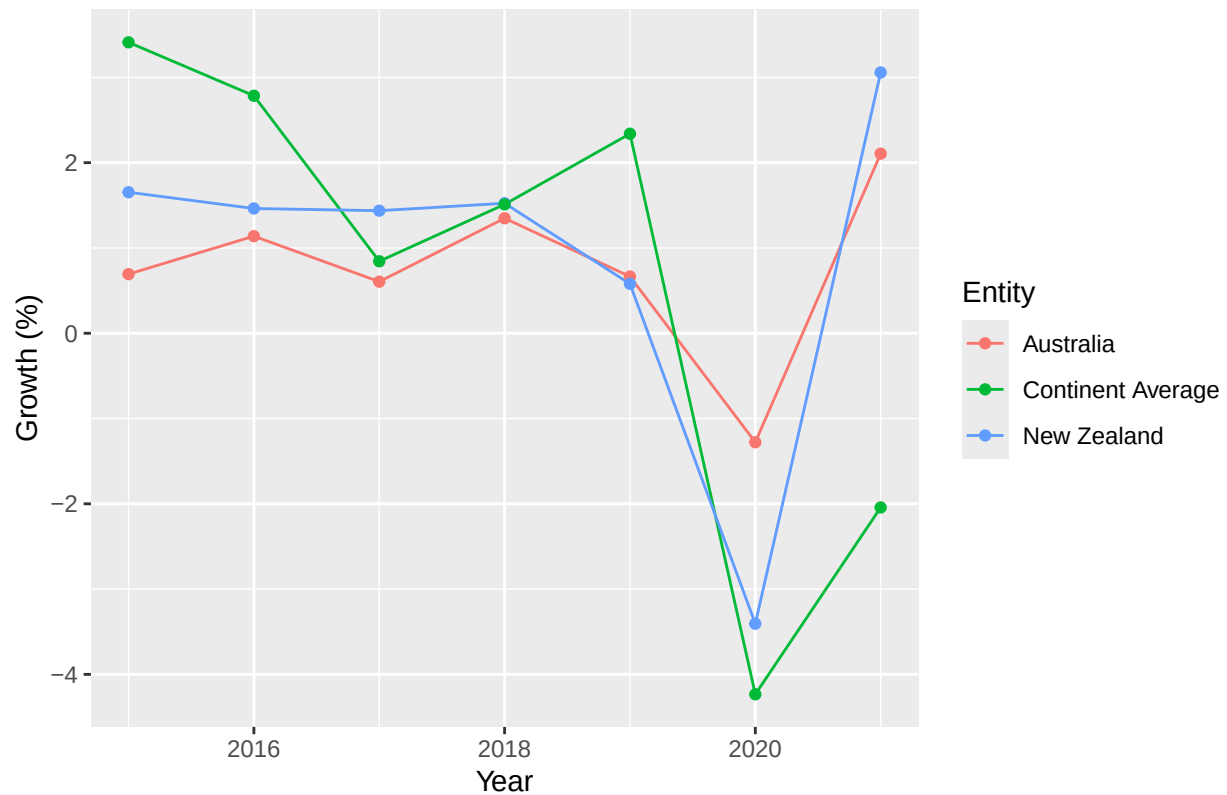


```
plot_drivers("South America")
```



```
plot_drivers("Oceania")
```

Top 2 GDP Growth vs Avg – Oceania



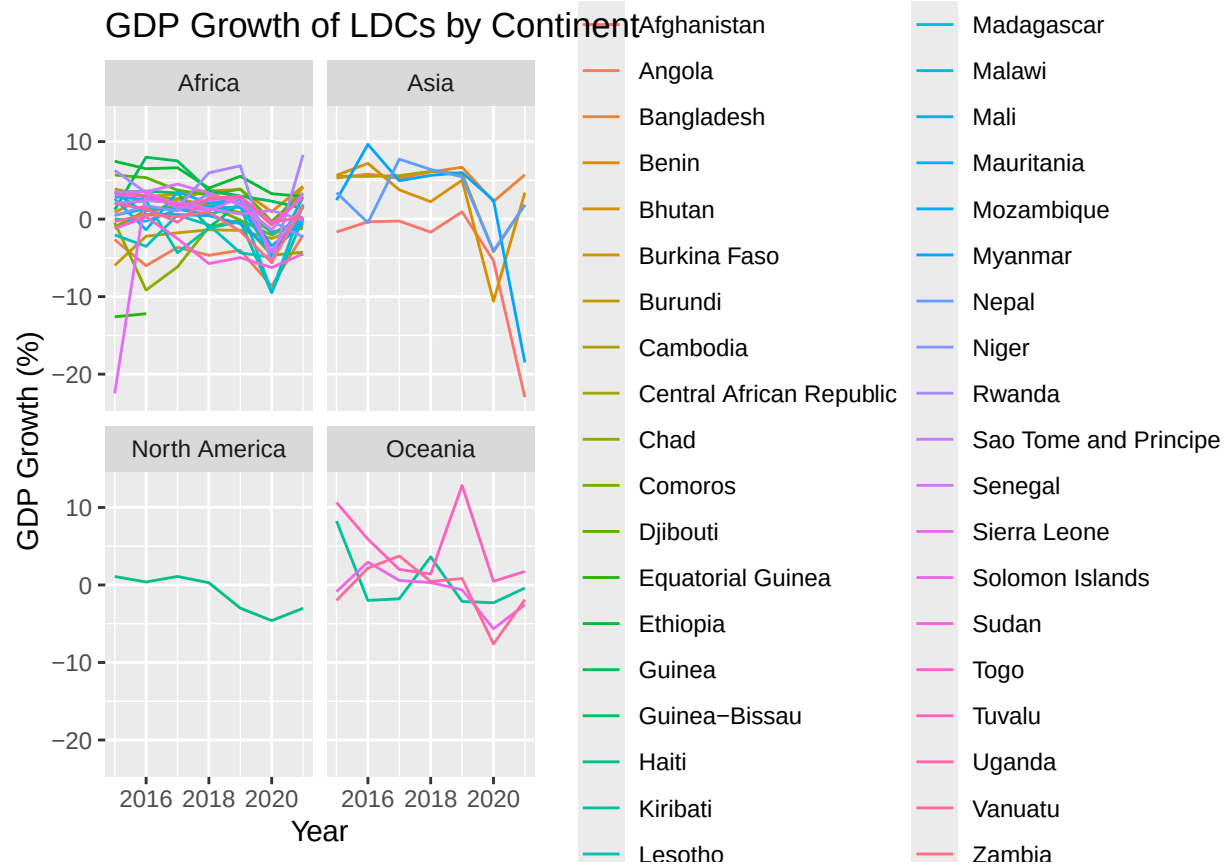
## Target 1.2 - LDC 7% Growth

Create an LDC database

```
ldc <- gdp_per_capita_df %>%
  filter (Ldc == TRUE)
```

Plot GDP growth by continent

```
ggplot(ldc, aes(x = Year, y = Growth, color = Entity)) +
  geom_line() +
  facet_wrap(~ Continent) +
  labs(
    title = "GDP Growth of LDCs by Continent",
    x = "Year",
    y = "GDP Growth (%)"
  )
```



Compute the weighted average GDP growth for each continent

```
ldc_weighted <- ldc %>%
  group_by(Year, Continent) %>%
  summarise(
    weighted_growth =
      sum(Gdp_per_capita * Growth, na.rm = TRUE) /
      sum(Gdp_per_capita, na.rm = TRUE),
    .groups = "drop"
  )
```

Plot weighted GDP growth by continent with a 7% reference line

```
ggplot(ldc_weighted, aes(x = Year, y = weighted_growth, color = Continent)) +
  geom_point() +
  geom_line() +
  facet_wrap(~ Continent, scales = "free_y") +
  geom_hline(yintercept = 7, linetype = "dashed", color = "red") +
  labs(
    title = "Weighted GDP Growth of LDCs by Continent",
    x = "Year",
    y = "Weighted GDP Growth (%)"
  )
```

## Weighted GDP Growth of LDCs by Continent



Identify which LDCs have had a growth that reached 7% and plot their growth

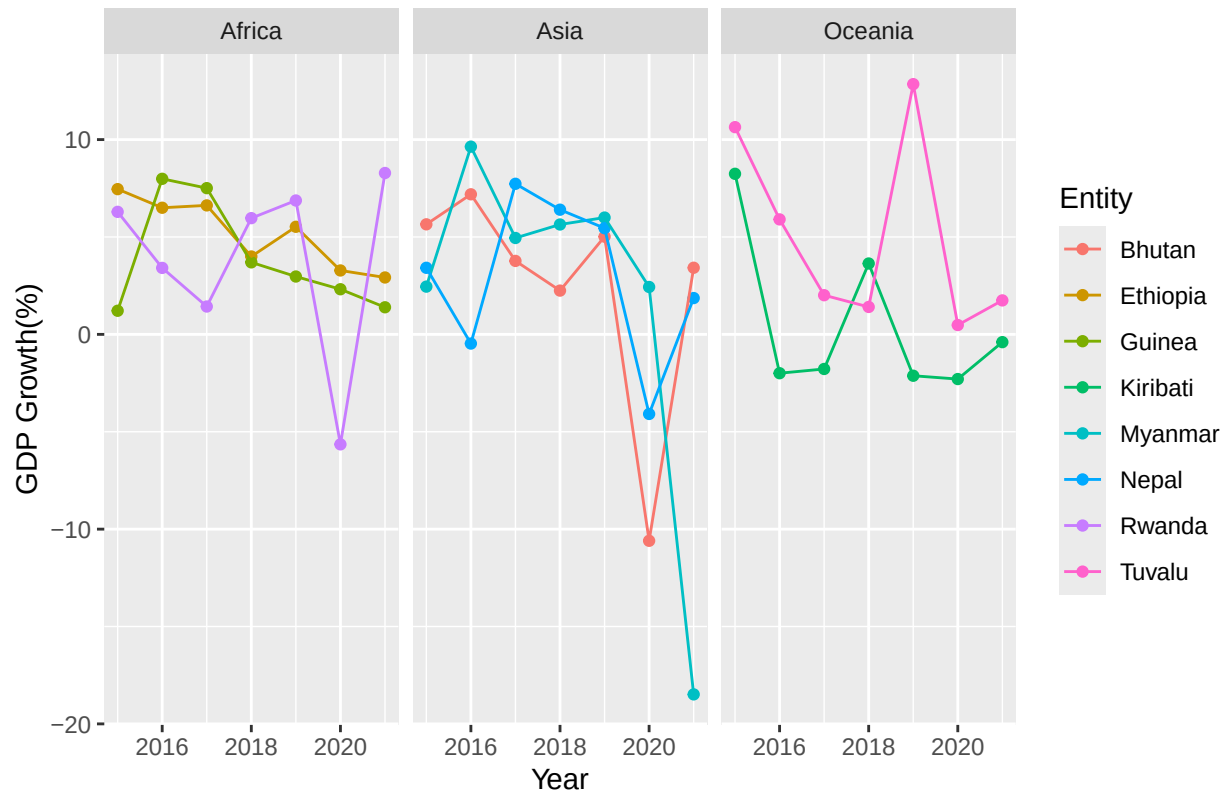
```
ldc_7 <- ldc %>%
  group_by(Entity) %>%
  filter(any(Growth >= 7)) %>%
  ungroup()
```

Plot their graph

```
ggplot(ldc_7, aes(x = Year, y = Growth, color = Entity)) +
  geom_point() +
  geom_line() +
  facet_wrap(~ Continent) +
  labs(
    title = "GDP Growth of LDCs that reached 7%",
    x = "Year",
    y = "GDP Growth(%)"
  )
```



## GDP Growth of LDCs that reached 7%



Determine which countries have had the highest growth over the years

```
ldc %>%
  group_by(Entity) %>%
  summarise(avg_growth = mean(Growth, na.rm = TRUE)) %>%
  arrange(desc(avg_growth))
```

```
## # A tibble: 39 x 2
##   Entity      avg_growth
##   <chr>      <dbl>
## 1 Bangladesh    5.30
## 2 Ethiopia      5.19
## 3 Tuvalu        5.00
## 4 Guinea        3.87
## 5 Rwanda        3.80
## 6 Cambodia      3.75
## 7 Djibouti      3.54
## 8 Nepal         2.90
## 9 Senegal       2.67
## 10 Burkina Faso  2.41
## # i 29 more rows
```

REF: 28/11/2025 - asked ChatGPT - "i have text near the last points on the x axis, how can i make it so it doesn't overlap the axis" 28/11/2025 - asked ChatGPT - "I have a facet plot in ggplot with some continents having tons of lines and some only one. How can I adjust the y-axis scales so each graph uses the right space for its lines and they don't look squished or empty?"